

BLUE REPORT

Testosterone replacement therapy can be cautiously considered in this patient with Grade Group 3 prostate cancer after radical prostatectomy, given his favorable pathology (negative margins, no metastatic disease) and persistently undetectable PSA. While the Endocrine Society guideline states there is "inadequate evidence to quantify the risk-benefit ratio" and lacks a general recommendation, the **American Urological Association** and **NCCN Survivorship guidelines** acknowledge that TRT may be considered in selected prostate cancer survivors with no evidence of recurrent disease, coordinated with the oncology team.[\[1\]\[2\]](#)

Oncologic Safety

The available evidence, though limited to retrospective studies, is reassuring. A 2025 systematic review found **biochemical recurrence rates of 0-7% post-prostatectomy** in TRT-exposed cohorts with up to 60 months of follow-up, with no studies showing increased recurrence risk compared to controls.[\[3\]](#) A large 2025 study from Memorial Sloan Kettering including 5,199 men (198 receiving TRT) with Grade Groups 1-3 found a **nonsignificantly decreased risk of biochemical recurrence** (HR 0.84, 95% CI 0.48-1.46) with TRT, and 5-year BCR probability <2% in both groups.[\[4\]](#) Multiple systematic reviews and meta-analyses consistently report no increased risk of biochemical recurrence with TRT after definitive treatment.[\[5\]\[3\]\[6\]\[7\]](#)

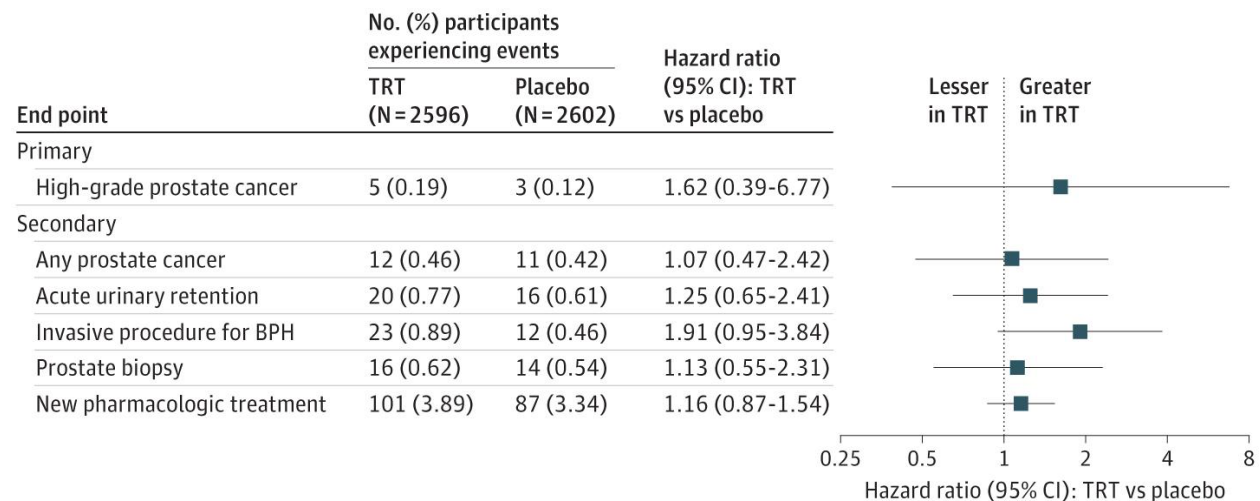


Figure 2 Incidence of Primary (High-Grade Prostate Cancer) and Secondary Prostate Safety End Points.

[Prostate Safety Events During Testosterone Replacement Therapy in Men With Hypogonadism: A Randomized Clinical Trial](#). JAMA Netw Open. November 30, 2023.

The following figure from the TRAVERSE trial demonstrates that testosterone treatment did not increase the risk of high-grade prostate cancer or other prostate safety endpoints in men with hypogonadism, though men at high risk for prostate cancer were excluded from this trial.

However, important caveats exist: **all studies are retrospective with small sample sizes, short follow-up duration, and lack long-term survival data.**[\[3\]](#)[\[5\]](#) The Endocrine Society notes that existing trials are "underpowered and of too short duration" to detect effects attributable to testosterone therapy.[\[9\]](#) Additionally, TRT increases PSA levels, which may lead to increased biopsies and detection of subclinical cancers.[\[10\]](#)

Patient Selection Criteria

This patient represents a **favorable candidate** based on published selection criteria:[\[2\]](#)[\[11\]](#)[\[12\]](#)

- Organ-confined disease (no seminal vesicle invasion, negative lymph nodes)
- Negative surgical margins
- Undetectable PSA postoperatively (maintained for 12 months)
- Grade Group 3 (intermediate risk, not high-risk)
- Confirmed biochemical hypogonadism on two morning measurements

The AUA guideline specifically states TRT "can be considered in men who have undergone radical prostatectomy with favorable pathology (e.g., negative margins, negative seminal vesicles, negative lymph nodes), and who have undetectable PSA postoperatively."[\[2\]](#)

Expected Symptomatic Benefits

TRT consistently improves hypogonadal symptoms in prostate cancer survivors:[\[3\]](#)[\[13\]](#)

- **Sexual function:** Improved libido, erectile function (though less effective if erectile dysfunction is the primary complaint)[\[10\]](#)[\[14\]](#)
- **Energy and fatigue:** Significant improvements reported across studies[\[13\]](#)
- **Mood:** Decreased depressive symptoms[\[10\]](#)[\[13\]](#)
- **Anemia correction:** In 42.9% of anemic patients in one cohort[\[13\]](#)
- **Metabolic effects:** Potential improvements in body composition, though data are mixed[\[13\]](#)

Testosterone levels typically increase from hypogonadal range (<300 ng/dL) to mid-normal range (300-500 ng/dL) with treatment.[\[4\]](#)[\[13\]](#)

Monitoring Protocol

If TRT is initiated, rigorous surveillance is essential:[\[2\]](#)[\[5\]](#)[\[12\]](#)

Baseline assessment:

- Confirm hypogonadism with repeat morning total and free testosterone, LH, prolactin
- Document undetectable PSA
- Digital rectal examination

Ongoing monitoring:

- PSA every 3-6 months initially, then every 6-12 months[\[2\]](#)[\[15\]](#)
- Testosterone levels to ensure therapeutic range
- Hematocrit (risk of polycythemia)
- Clinical assessment for lower urinary tract symptoms

Thresholds for concern:

- PSA ≥ 0.2 ng/mL with confirmatory rise (biochemical recurrence definition)[\[4\]](#)
- Rapid PSA velocity
- Abnormal digital rectal examination

Shared Decision-Making Discussion

The decision requires **informed consent** addressing:[\[1\]](#)[\[9\]](#)[\[2\]](#)

- **Limited evidence:** No randomized controlled trials; all data from retrospective studies with short follow-up
- **Potential risks:** Possible stimulation of occult disease, increased PSA leading to additional

testing/biopsies, general TRT risks (polycythemia, cardiovascular events including pulmonary embolism)[10]

- **Uncertain long-term safety:** Lack of survival outcome data
- **Need for intensive monitoring:** Lifelong PSA surveillance required
- **Coordination with oncology team:** NCCN recommends coordination with surgeon/oncologist[1]

The NCCN Survivorship guidelines note that "when to initiate treatment for low testosterone in prostate cancer survivors...is controversial and should be coordinated with the patient's PCP (ie, surgeon, oncologist, radiation oncologist)."[1]

Recommendation

Given this patient's favorable pathology, persistently undetectable PSA, confirmed hypogonadism, and significant symptoms affecting quality of life, **TRT may be cautiously offered** with the understanding that this represents an individualized decision based on limited evidence. The patient should be counseled that while retrospective data are reassuring, definitive safety data from prospective trials are lacking. If he chooses to proceed, establish a rigorous PSA monitoring protocol and maintain close coordination with his urologic oncologist.

Would you like me to outline specific TRT formulation options and dosing strategies for this patient, or discuss how to counsel him if PSA becomes detectable during treatment?

References

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